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Assignment 2 — Individual

DNS Spoofing Attack Simulation and Incident Response Plan

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ABSTRACT

This report details a simulation of a DNS spoofing attack in a VMware lab environment as part of a Security Operations Centre (SOC) exercise. The attack was simulated using Ettercap 0.8.3.1 on Kali Linux as the attacker and a Windows host as the victim, using ARP cache poisoning as the initial Man-in-the-Middle attack vector and the dns_spoof plugin to spoof DNS responses and redirect the victim to a fake website. Forensic evidence was captured using Wireshark and compared to the Windows ARP cache, revealing a spoofed DNS A record and duplicate MAC addresses as the Indicators of Compromise. An Incident Response Plan was created following the guidelines of NIST Special Publication 800-61 Revision 2, including preparation, detection and analysis, containment, eradication, recovery and post-incident activities. The exercise shows that DNS spoofing can be carried out using common penetration testing tools on an insecure local network, and that its indicators can be detected via network packet capture.

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1. Introduction

Simulating cyber-attacks is an essential component of SOC training - it allows security analysts to experiment with attack methods, gather forensic evidence, and develop response processes before they are confronted with the same attacks in the wild. This report outlines one such exercise, with four tasks: choosing and researching a real attack, performing the attack in a lab environment, gathering forensic evidence from the attack, and creating an Incident Response Plan based on the evidence.

The attack type selected is DNS Spoofing and it was performed in an isolated VMware lab environment with Kali Linux (192.168.61.131) as the attacker and a Windows host (192.168.61.133) as the victim. The ARP poisoning (initial MiTM) and DNS response forgery (to redirect to a cloned APU login page) were carried out using Ettercap 0.8.3.1. Wireshark and the Windows ARP cache were used to gather forensic evidence. The Incident Response Plan in Task 4 is based on the NIST SP 800-61 Revision 2 guidelines.

2. Task 1: Selection of Cyber-Attack

2.1 Attack Concept

The Domain Name System (DNS) translates domain names into IP addresses, but lacks inherent security. Dissanayake (2018) explains that DNS cache poisoning is a type of attack that redirects traffic by injecting a spoofed IP address into the cache of a resolver - all clients that use that resolver are then unwittingly redirected. Chandrasekaran et al. (2023) explain this is used to spread malware, launch phishing attacks or redirect users to malicious servers. The Kaminsky attack in 2008 demonstrated a resolver could be poisoned in a matter of seconds, prompting a re-evaluation of DNS security (Dissanayake, 2018). For the simulation, the APU student portal (apspace.apu.edu.my) is the target, redirected to a fake login page hosted on Kali's Apache web server.

2.2 Techniques Involved

The first step in spoofing DNS on a LAN is to poison ARP. ARP is stateless - there's no authentication and a host will accept and cache an ARP reply even if it didn't request it. As Srikar Vishnu Datta et al. (2022) explained, "The man-in-the-middle attack is the most common attack that takes advantage of the vulnerability that exists in the ARP protocol." The attacker floods the network with forged ARP replies claiming the gateway's IP belongs to the attacker's MAC. The victim's ARP table is poisoned, and the victim's traffic destined for the gateway is redirected via the attacker's PC.

Once the network traffic is redirected, DNS spoofing follows naturally. The attacker sees the victim's DNS request, and responds with the right domain, but a wrong IP. The victim's OS caches it, and the redirect is done. Srikar Vishnu Datta et al. (2022) confirm this is how it works: the standard extension of MITM is to forge the DNS response, and the user will be redirected to a spoofed page.

2.3 Tools Commonly Used

The primary one is Ettercap. It is installed by default on Kali and performs ARP poisoning and DNS spoofing (via the dns_spoof plugin). The plugin loads the configuration file (etter.dns) in which the attacker specifies the domains and the IP to which they should be redirected. The attacker's computer hosts the malicious login page via Apache2. Wireshark logs the network traffic

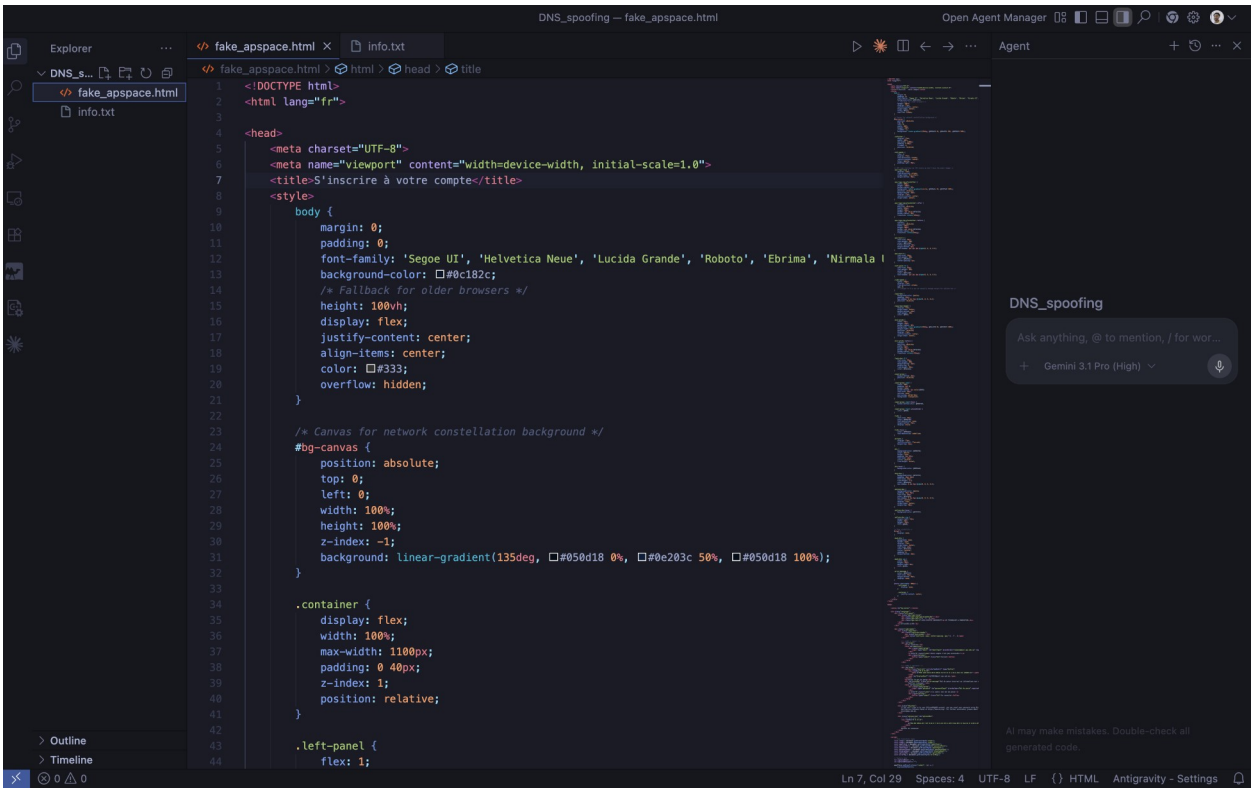
and is used to produce the forensic evidence in Task 3. This trio of tools is similar to the one used in other attack studies (Srikar Vishnu Datta et al., 2022).

3. Task 2: Cyber-Attack Simulation

The attack ran on an isolated VMware network (192.168.61.0/24). Kali Linux at 192.168.61.131 was the attacker, Windows at 192.168.61.133 was the victim, and the gateway sat at 192.168.61.2. Kali had Apache2 running and the cloned APSpace page as the index. The steps are illustrated with screenshots.

Step 1: Preparation of the Malicious Web Page

Before the attack, a replica of the APU APSpace login page was crafted in HTML and placed in the document root of the Apache web server (/var/www/html/index.html). It was verified by visiting http://localhost on the Kali machine. This sets up the fake login page that the victim will be redirected to when the DNS spoofing succeeds.




```
<!DOCTYPE html>
<html lang="fr">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>S'inscrire à votre compte</title>
    <style>
        body {
            margin: 0;
            padding: 0;
            font-family: 'Segoe UI', 'Helvetica Neue', 'Lucida Grande', 'Roboto', 'Ebrima', 'Nirmala
            background-color: #0c182c;
            /* Fallback for older browsers */
            height: 100vh;
            display: flex;
            justify-content: center;
            align-items: center;
            color: #333;
            overflow: hidden;
        }

        /* Canvas for network constellation background */
        #bg-canvas {
            position: absolute;
            top: 0;
            left: 0;
            width: 100%;
            height: 100%;
            z-index: -1;
            background: linear-gradient(135deg, #050d18 0%, #0e203c 50%, #050d18 100%);
        }

        .container {
            display: flex;
            width: 100%;
            max-width: 1100px;
            padding: 0 40px;
            z-index: 1;
            position: relative;
        }

        .left-panel {
            flex: 1;
```

Figure 1 – Fake APSpace login page HTML source code prepared on the attacker machine

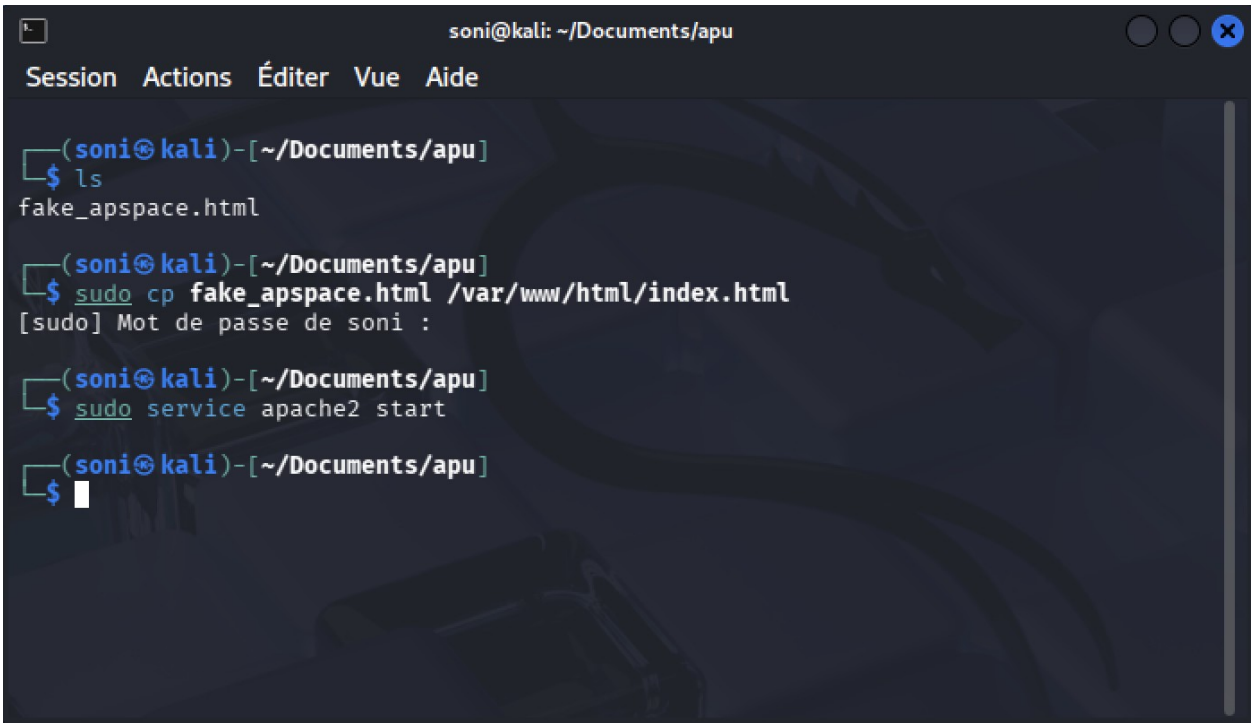


Figure 2 – Terminal showing file copy to /var/www/html and Apache service start commands

Step 2: Enabling IP Forwarding

The sysctl command was used to enable IP forwarding on the Kali machine to ensure that intercepted traffic is relayed to its destination without disrupting the victim's connectivity, keeping the MiTM position transparent.

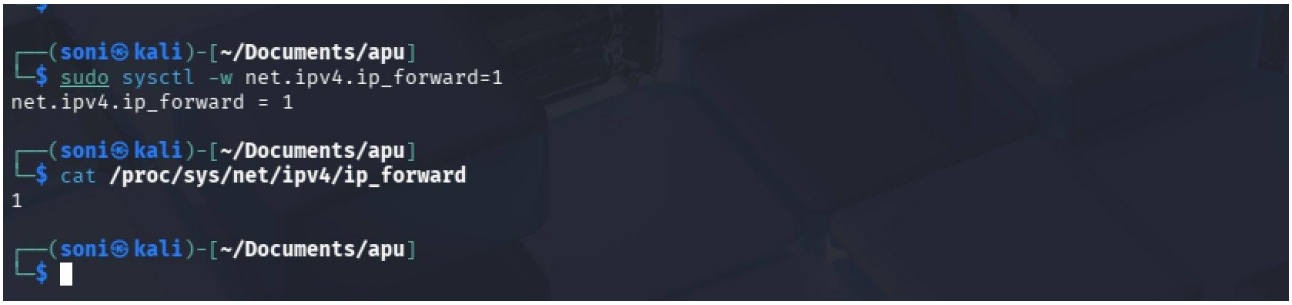


Figure 3 – IP forwarding enabled: net.ipv4.ip_forward = 1 confirmed

Step 3: Configuring the DNS Spoof Rules

The Ettercap DNS plugin configuration file (/etc/ettercap/etter.dns) was opened in nano. Two A records were added at the bottom of the file, pointing both the apex domain (apspace.apu.edu.my) and its wildcard subdomain (*.apspace.apu.edu.my) to the Kali machine's IP (192.168.61.131). This ensures any DNS query for the target domain, with or without a subdomain, resolves to the attacker's IP.

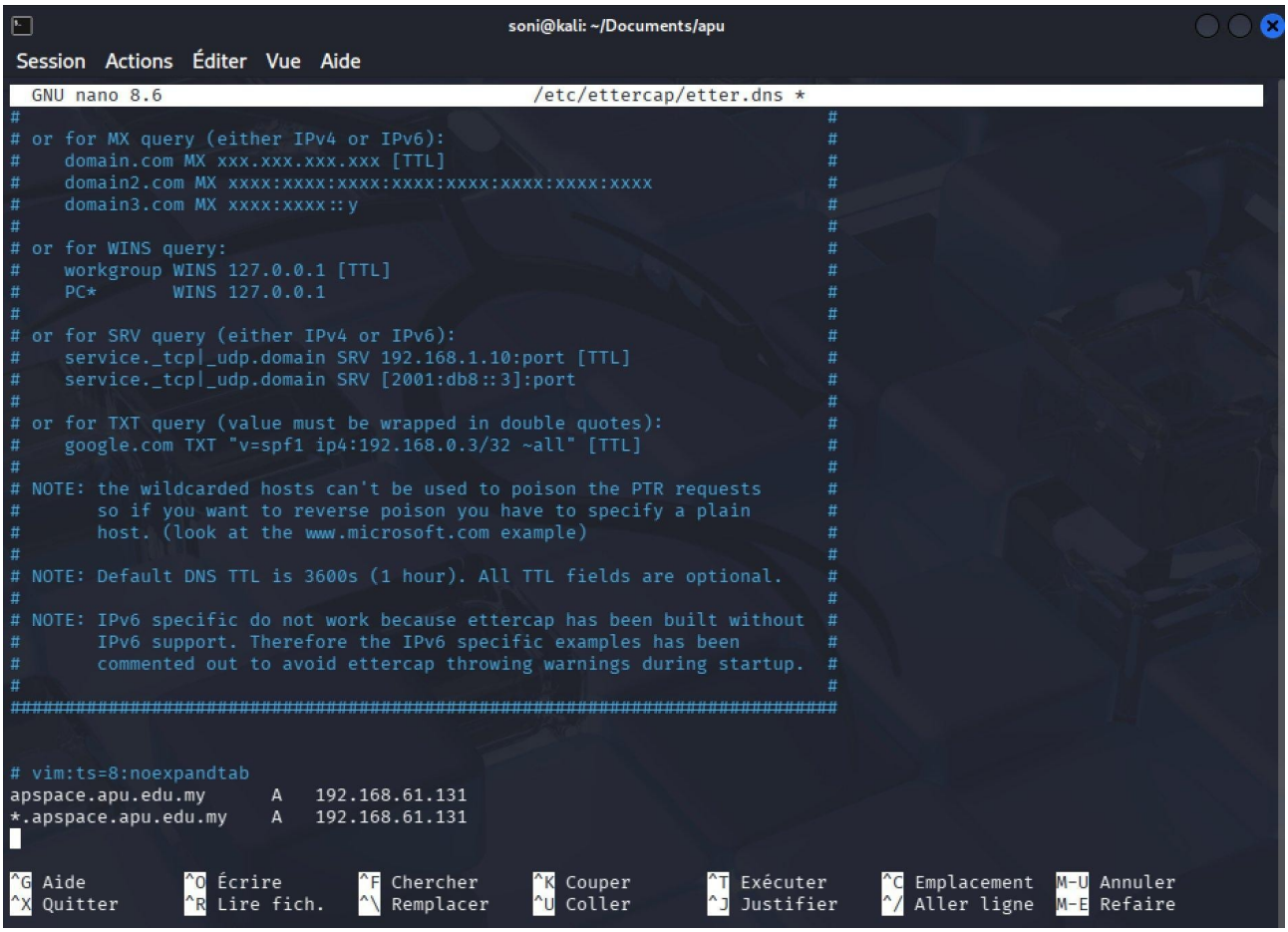


Figure 4 – etter.dns configuration showing spoofed A records pointing to 192.168.61.131

Step 4: Launching Ettercap

Ettercap was started in GUI mode with sudo ettercap -G. The setup screen confirmed eth0 as the primary interface with unified sniffing enabled on startup.

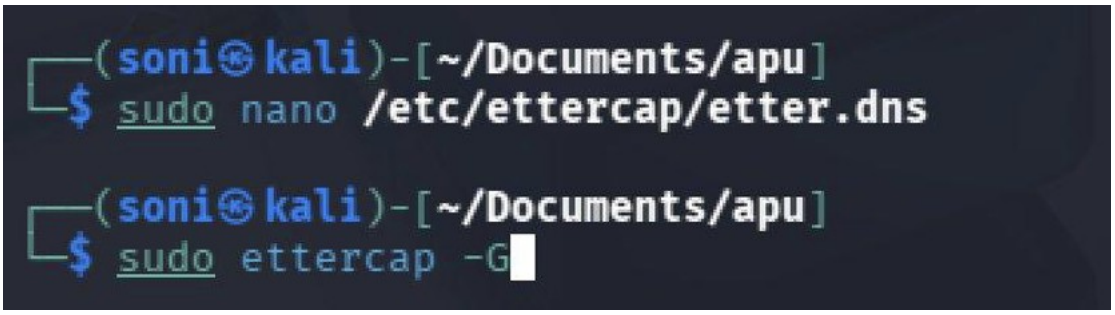


Figure 5 – Terminal showing the sudo ettercap -G command being issued

Figure 6 – Ettercap GUI setup screen confirming eth0 as the primary interface

Figure 7 – Ettercap started: listening on eth0 (192.168.61.131), Starting Unified sniffing confirmed

Step 5: Scanning for Hosts and Defining Targets

A host scan via the Hosts menu identified the gateway (192.168.61.2) and the victim Windows machine (192.168.61.133) with their MAC addresses. The gateway was added to Target 1 and the victim to Target 2, confirmed by the console log.

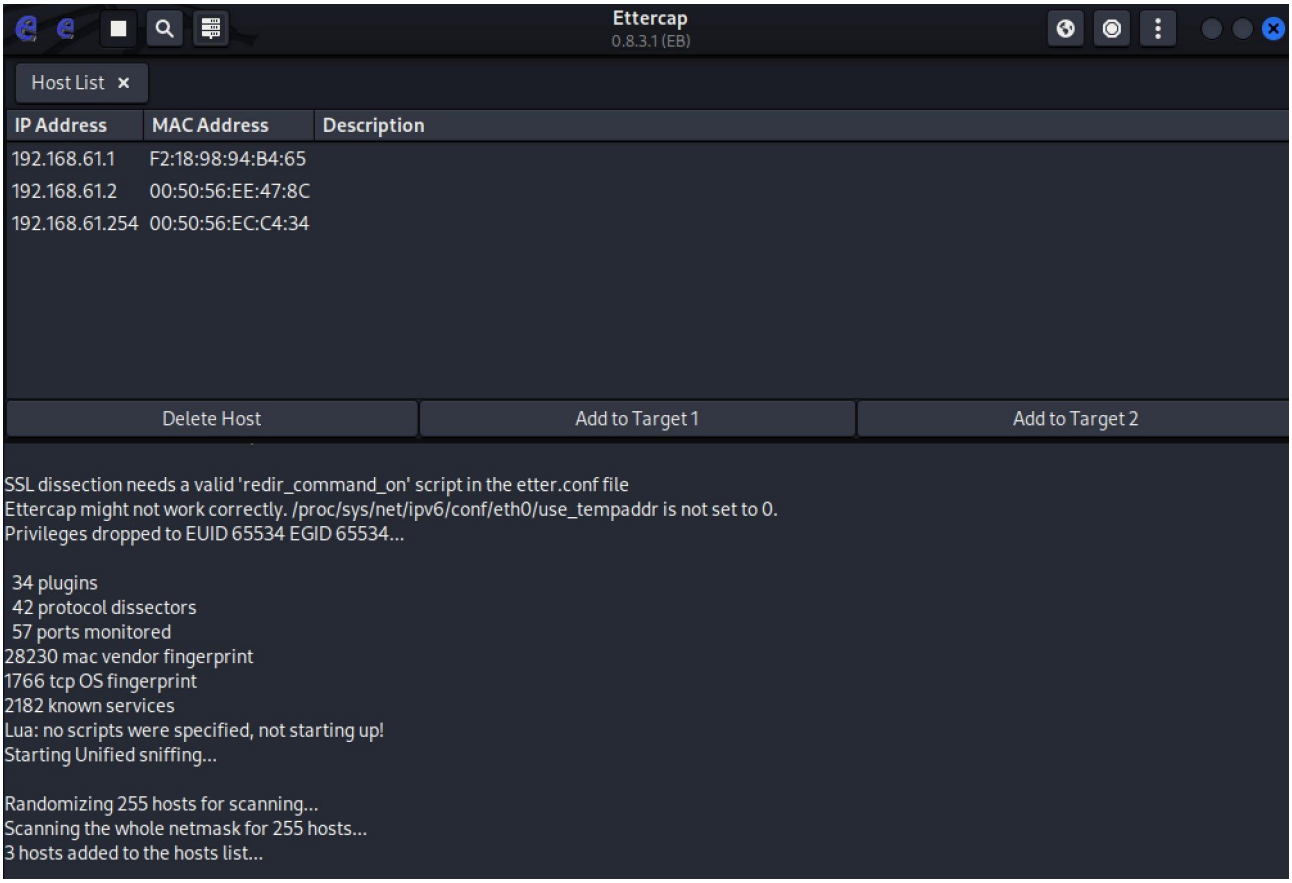


Figure 8 – Host scan completed: 3 hosts discovered on 192.168.61.0/24

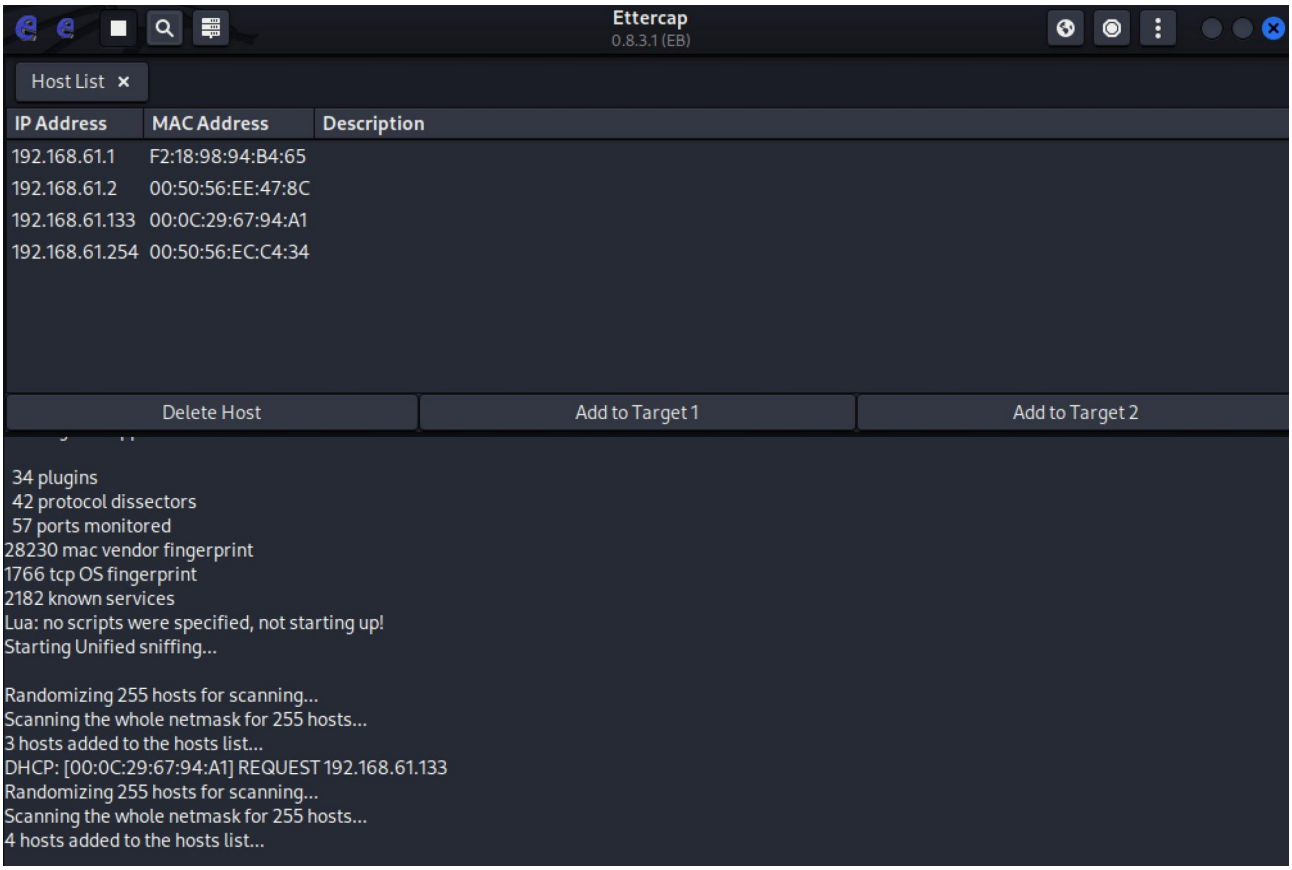


Figure 9 – Updated host list showing victim 192.168.61.133 detected via DHCP request

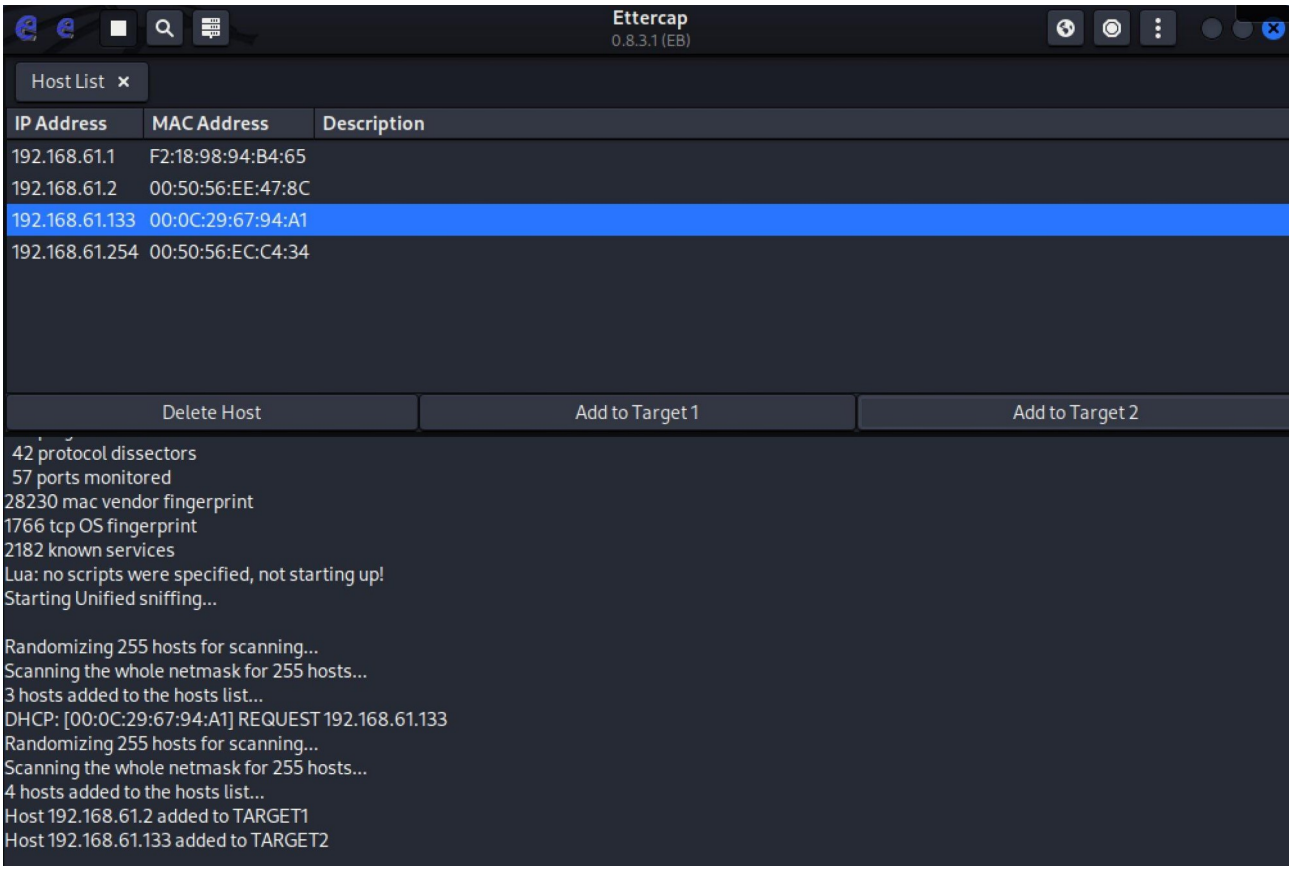


Figure 10 – Console log confirming 192.168.61.2 added to TARGET1 and 192.168.61.133 added to TARGET2

Step 6: Launching ARP Poisoning

ARP poisoning was started via MITM > ARP Poisoning, with 'Sniff remote connections' enabled. This causes Ettercap to send forged ARP replies to both the victim and the gateway, placing Kali in the middle of their traffic.

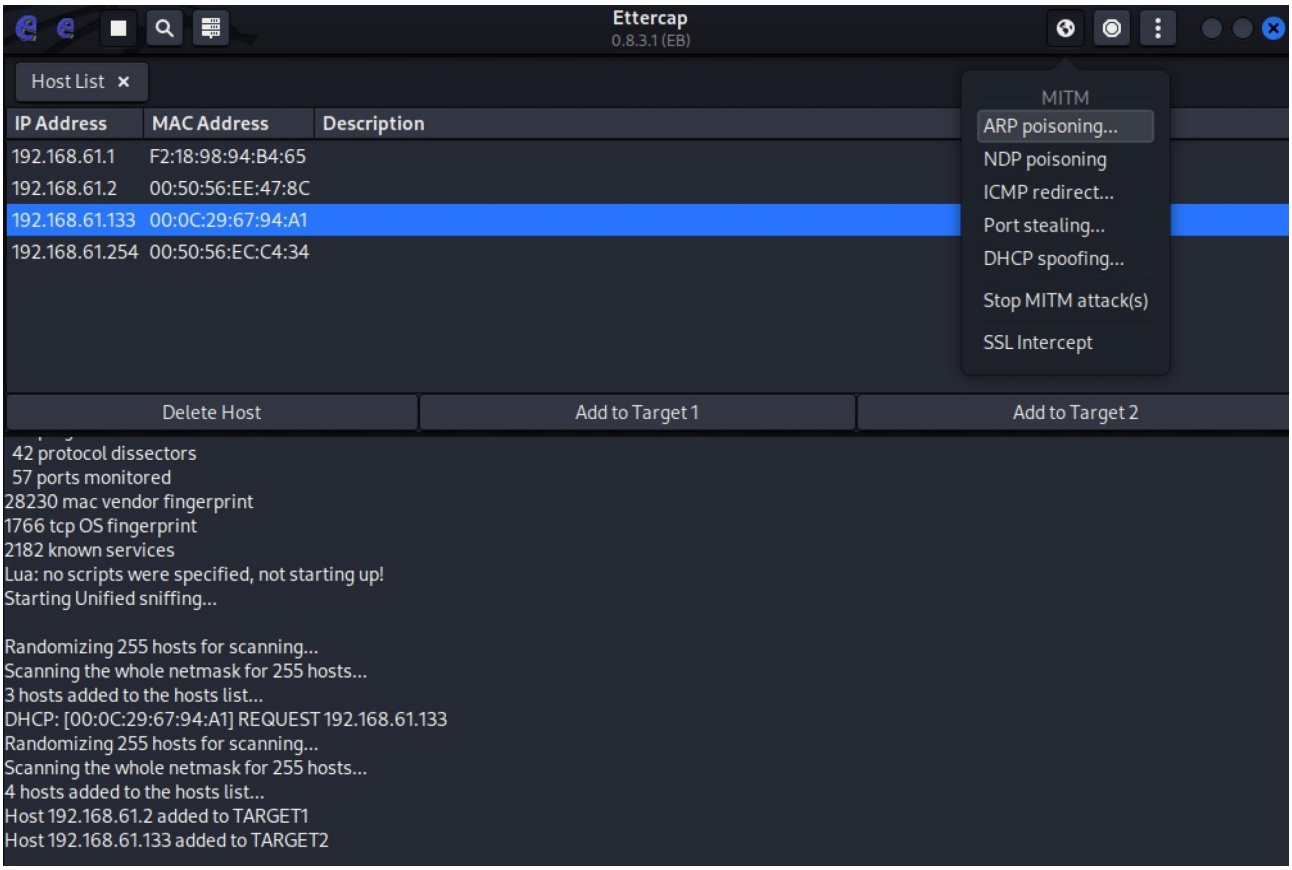


Figure 11 – MITM menu open with ARP Poisoning selected

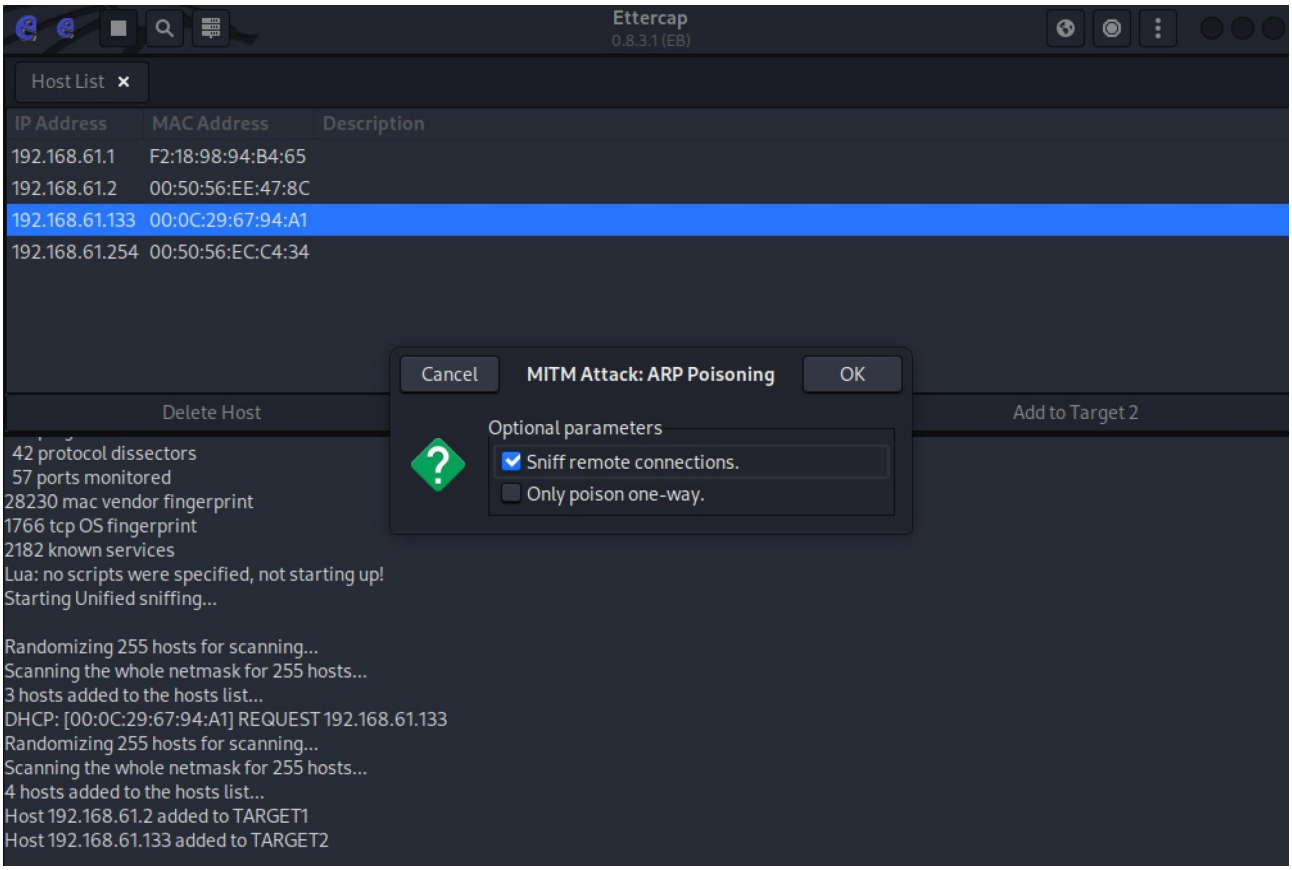


Figure 12 – ARP Poisoning dialog with 'Sniff remote connections' checked

Figure 13 – Console confirming ARP poisoning victims: GROUP 1 (gateway) and GROUP 2 (victim)

Step 7: Activating the dns_spoof Plugin

The dns_spoof plugin was activated by double-clicking its entry in Plugins > Manage Plugins. An asterisk (*) appeared next to the plugin name confirming activation, and the console logged 'Activating dns_spoof plugin...'.

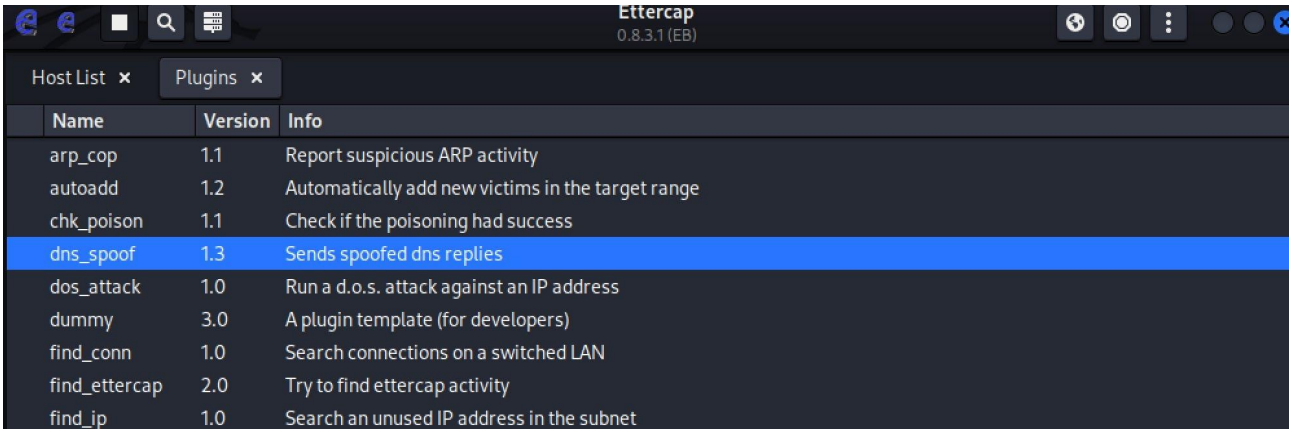


Figure 14 – Plugin list showing dns_spoof (version 1.3) selected, described as 'Sends spoofed dns replies'

Figure 15 – Asterisk (*) next to dns_spoof confirming activation; console shows 'Activating dns_spoof plugin...'

Step 8: Confirming the DNS Spoof and Observing the Result

The victim's browser was directed to http://apspace.apu.edu.my. Instead of the real APU portal, it loaded the cloned login page from Kali's Apache server. The address bar showed the target domain with a "Not secure" warning, confirming the DNS response was forged and the redirect successful. The Ettercap console logged the spoof event, showing the A record for apspace.apu.edu.my was spoofed to 192.168.61.131 with a TTL of 3600 seconds.

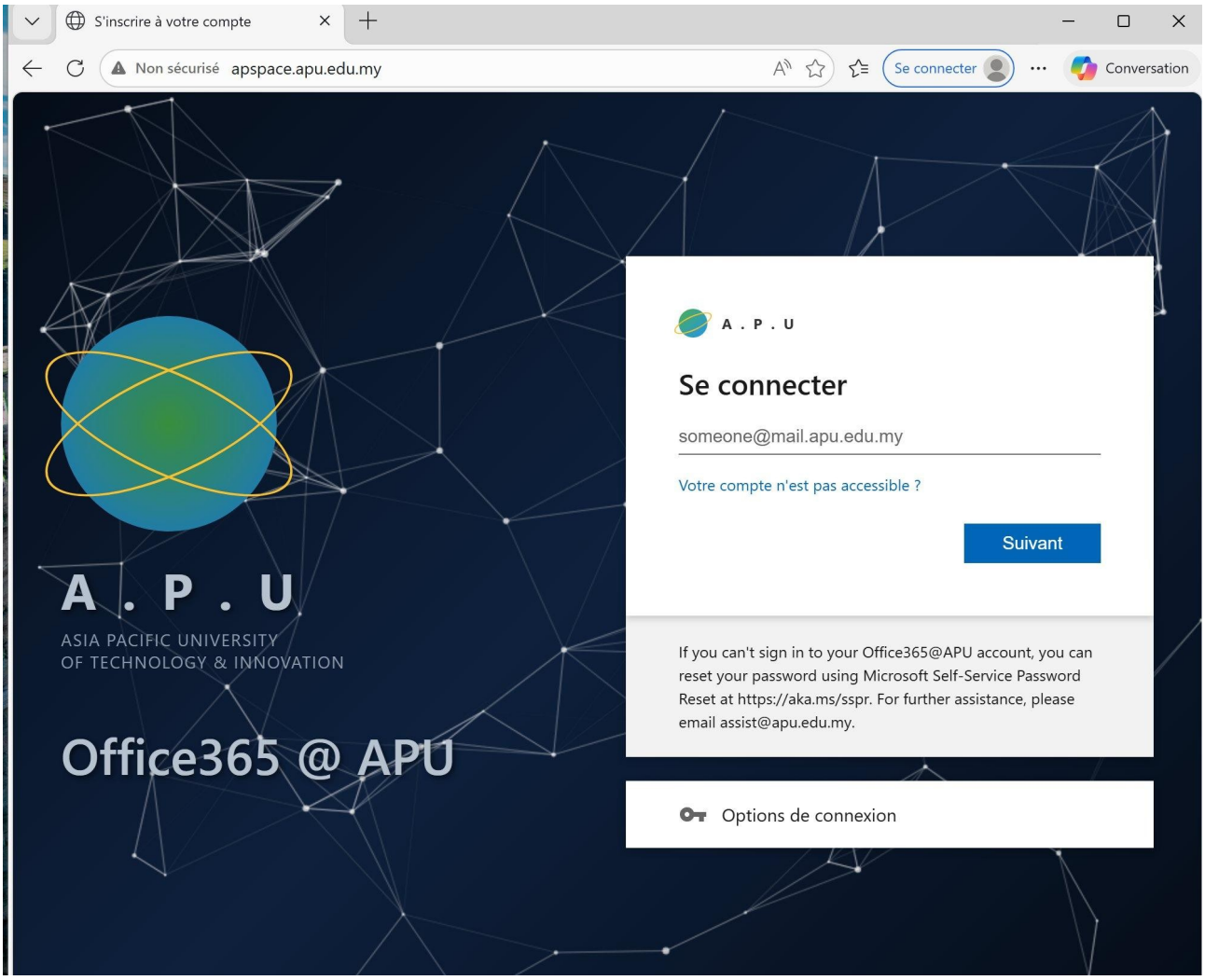


Figure 16 – Windows victim browser displaying the cloned APSpace login page at apspace.apu.edu.my ('Not secure' indicator visible)

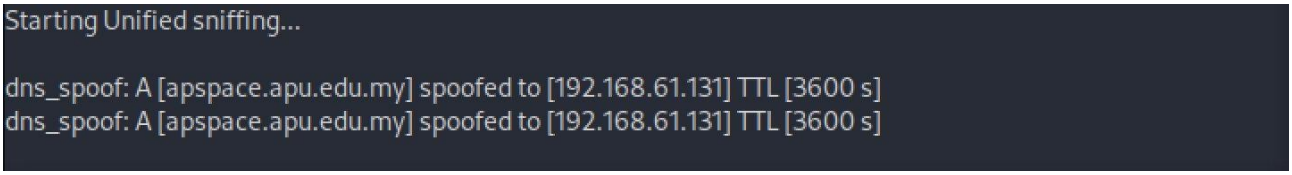


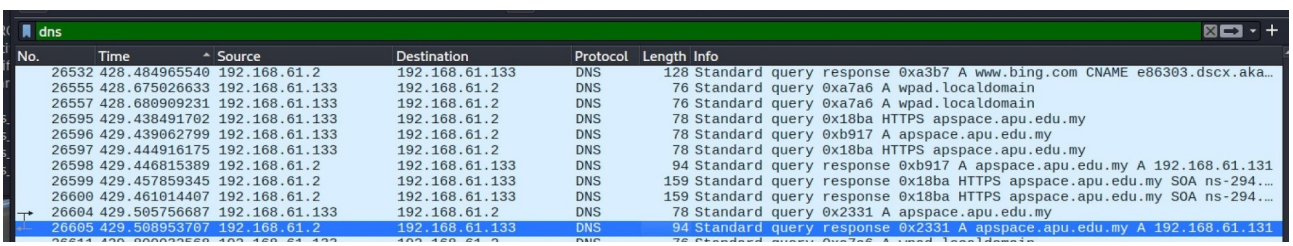
Figure 17 – Ettercap console confirming: dns_spoof: A [apspace.apu.edu.my] spoofed to [192.168.61.131] TTL [3600 s]

4. Task 3: Capture and Analysis of Forensic Evidence

4.1 IoC: Forged DNS Response Captured in Wireshark

Wireshark ran throughout the simulation with a "dns" display filter active. The capture provides the primary Indicator of Compromise. Wijayanto et al. (2023) confirm that the key forensic data in ARP/DNS spoofing cases — attacker IP, attacker MAC, victim IP, victim MAC, and timestamps — is recoverable from PCAP files using Wireshark.

Figure 18 shows the DNS traffic when the victim's browser queried apspace.apu.edu.my. Packets 26596 and 26599 are the legitimate A and HTTPS queries from the victim (192.168.61.133) to the resolver (192.168.61.2). The critical entry is packet 26605: a Standard query response returning an A record of 192.168.61.131 — the Kali attacker — instead of APU's real server. That discrepancy is the IoC. The IP in the Wireshark capture matches the etter.dns configuration, and the timestamp aligns with the dns_spoof log in Figure 17, establishing a clear forensic link.



dns

No.	Time	Source	Destination	Protocol	Length	Info
26532	428.484965540	192.168.61.2	192.168.61.133	DNS	128	Standard query response 0xa3b7 A www.bing.com CNAME e86303.dscx.aka…
26555	428.675026633	192.168.61.133	192.168.61.2	DNS	76	Standard query 0xa7a6 A wpad.localdomain
26557	428.680909231	192.168.61.133	192.168.61.2	DNS	76	Standard query 0xa7a6 A wpad.localdomain
26595	429.438491702	192.168.61.133	192.168.61.2	DNS	78	Standard query 0x18ba HTTPS apspace.apu.edu.my
26596	429.439062799	192.168.61.133	192.168.61.2	DNS	78	Standard query 0xb917 A apspace.apu.edu.my
26597	429.444916175	192.168.61.133	192.168.61.2	DNS	78	Standard query 0x18ba HTTPS apspace.apu.edu.my
26598	429.446815389	192.168.61.2	192.168.61.133	DNS	94	Standard query response 0xb917 A apspace.apu.edu.my A 192.168.61.131
26599	429.457859345	192.168.61.2	192.168.61.133	DNS	159	Standard query response 0x18ba HTTPS apspace.apu.edu.my SOA ns-294.…
26600	429.461014407	192.168.61.2	192.168.61.133	DNS	159	Standard query response 0x18ba HTTPS apspace.apu.edu.my SOA ns-294.…
26604	429.505756687	192.168.61.133	192.168.61.2	DNS	78	Standard query 0x2331 A apspace.apu.edu.my
26605	429.508953707	192.168.61.2	192.168.61.133	DNS	94	Standard query response 0x2331 A apspace.apu.edu.my A 192.168.61.131

Figure 18 – Wireshark capture (dns filter) showing forged A record response: apspace.apu.edu.my resolved to 192.168.61.131 (attacker) instead of the legitimate IP

4.2 Supporting Evidence: ARP Cache Poisoning on the Victim

The Windows ARP cache was checked with arp -a during the attack. Figure 19 shows both 192.168.61.2 (the gateway) and 192.168.61.131 (the attacker) mapped to the same MAC address: 00-0c-29-c0-a1-34. This should never happen — each IP has a unique MAC in a healthy network. Two IPs sharing one MAC is the signature of ARP cache poisoning, confirming the MiTM position was in place before the DNS spoof ran. Wijayanto et al. (2023) report that Kaspersky's 2021 Incident Response data shows ARP spoofing accounted for 22% of all suspicious network activity investigations that year.

```
PS C:\Users\sondi> arp -a

Interface : 192.168.61.133 --- 0xb
  Adresse Internet      Adresse physique      Type
  192.168.61.2          00-0c-29-c0-a1-34     dynamique
  192.168.61.131        00-0c-29-c0-a1-34     dynamique
  192.168.61.254        00-50-56-ec-c4-34     dynamique
  192.168.61.255        ff-ff-ff-ff-ff-ff     statique
  224.0.0.22            01-00-5e-00-00-16     statique
  224.0.0.251           01-00-5e-00-00-fb     statique
  224.0.0.252           01-00-5e-00-00-fc     statique
  239.255.255.250       01-00-5e-7f-ff-fa     statique
  255.255.255.255       ff-ff-ff-ff-ff-ff     statique
PS C:\Users\sondi>
```

Figure 19 – Windows arp -a output: 192.168.61.2 (gateway) and 192.168.61.131 (attacker) share MAC 00-0c-29-c0-a1-34, confirming ARP cache poisoning

5. Task 4: Incident Response Plan

This Incident Response Plan follows the NIST Special Publication 800-61 Revision 2, Computer Security Incident Handling Guide (Cichonski et al., 2012) which outlines the four phases of the incident response process: Preparation, Detection and Analysis, Containment, Eradication and Recovery, and Post-Incident Activity. As NIST SP 800-61 defines, "computer security incident response has become an important component of information technology (IT) programs", and the guidelines are intended to be applicable regardless of the hardware platform, operating system or application - making it relevant to the DNS spoofing scenario in the report.

5.1 Phase 1 — Preparation

The organisation should have a CSIRT and escalation procedures. In the case of DNS spoofing, network monitoring should identify unusual ARP traffic and unexpected DNS responses. People should be aware that certificate errors and "Not secure" warnings can be the result of spoofing, not just configuration errors. Dynamic ARP Inspection (DAI) should be configured on controlled switches, and a list of expected ARP mappings maintained for key hosts. Chandrasekaran et al. (2023) list DNSSEC as the long-term control, along with source port randomisation and response rate limiting.

5.2 Phase 2 — Detection and Analysis

The primary IoC is a DNS response with an IP address that differs from the organisation's authoritative DNS records - see Figure 18. IDS should raise an alarm on any unexpected IP response for a given domain. The second IoC is duplicate MACs in the ARP entry; a scheduled script running arp -a against baselines would detect the anomaly shown in Figure 19. When detected, the incident should be logged with a timestamp and priority. Wijayanto et al. (2023) report that analysis with Wireshark successfully recovers the attacker's IP and MAC addresses and the event timestamps required to classify the attack.

5.3 Phase 3 — Containment, Eradication, and Recovery

Containment: segment the affected area, MAC block the attacker at the access layer, and statically map the gateway and other important hosts. Eradication: clear the DNS cache on all affected DNS resolvers, and enable DNSSEC on authoritative zones and DAI on any switch that doesn't already have it. Chandrasekaran et al. (2023) list DNSSEC, source port randomisation and

response rate limiting as the three main solutions. Success is confirmed when the DNS resolution is correct from an external location and when the ARP traffic stops.

5.4 Phase 4 — Post-Incident Activity

A post-incident report should include the timeline, the ARP vector, the dns_spoof attack, IoCs, containment measures and time to recovery. If DNSSEC was not in place, it's the first item to be remedied. Training should include the "Not secure" warning in Figure 16. The NIST SP 800-61 (Cichonski et al., 2012) approach is simple: record the incident, learn from it, and prevent it.

6. Conclusion

The simulation involved the entire DNS spoofing attack chain - ARP poisoning for MiTM, dns_spoof to forge DNS responses, and a duplicate login page to redirect to. The attack was successful: the victim's browser redirected to the cloned APSpace page, with only a "Not secure" message to give it away. The forensic investigation in Task 3 linked the Ettercap log, Wireshark capture, and the Windows ARP cache to prove the attack. The IRP demonstrates that the clues from this attack - spoofed DNS A records and duplicate ARP entries - can be spotted with Wireshark and simple ARP monitoring. The problem is that they're not always set up. DNSSEC and Dynamic ARP Inspection fix the vulnerability; they are not difficult to set up.

7. References

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Wijayanto, A., Riadi, I., Prayudi, Y., & Sudinugraha, T. (2023). Network forensics against address resolution protocol spoofing attacks using trigger, acquire, analysis, report, action method. Register: Jurnal Ilmiah Teknologi Sistem Informasi, 8(2), 156–169. https://doi.org/10.26594/register.v8i2.2953